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Weakly Supervised Detection of Pheochromocytomas and Paragangliomas in CT using Noisy Data

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Abstract

Pheochromocytomas and Paragangliomas (PPGLs) are rare adrenal and extra-adrenal tumors that have metastatic potential. Management of patients with PPGLs mainly depends on the makeup of their genetic cluster: SDHx, VHL/EPAS1, kinase, and sporadic. CT is the preferred modality for precise localization of PPGLs, such that their metastatic progression can be assessed. However, the variable size, morphology, and appearance of these tumors in different anatomical regions can pose challenges for radiologists. Since radiologists must routinely track changes across patient visits, manual annotation of PPGLs is quite time-consuming and cumbersome to do across all axial slices in a CT volume. As such, PPGLs are only weakly annotated on axial slices by radiologists in the form of RECIST measurements. To ameliorate the manual effort spent by radiologists, we propose a method for the automated detection of PPGLs in CT via a proxy segmentation task. Weak 3D annotations (derived from 2D bounding boxes) were used to train both 2D and 3D nnUNet models to detect PPGLs via segmentation. We evaluated our approaches on an in-house dataset comprised of chest-abdomen-pelvis CTs of 255 patients with confirmed PPGLs. On a test set of 53 CT volumes, our 3D nnUNet model achieved a detection precision of 70% and sensitivity of 64.1%, and outperformed the 2D model that obtained a precision of 52.7% and sensitivity of 27.5% ($p < 0.05$). SDHx and sporadic genetic clusters achieved the highest precisions of 73.1% and 72.7% respectively. Our state-of-the-art findings highlight the promising nature of the challenging task of automated PPGL detection.

Keywords

CT; PPGL; Paraganglioma; Pheochromocytoma; Genetic clusters; Detection; Deep Learning

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¹Equal Contribution

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Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

1. Introduction

Pheochromocytomas and Paragangliomas (PPGLs) are rare neuroendocrine tumors that can appear sporadically in the body or due to the inheritance of various pathogenic variants. Pheochromocytomas originate from chromaffin cells in the adrenal medulla, and makeup 80–85% of PPGLs [1, 2]. Conversely, paragangliomas can originate from extra-adrenal chromaffin cells located in the head and neck, chest, abdomen or pelvis regions, and only comprise 15–20% of PPGLs [1, 2]. PPGLs are mostly found in the neck, abdomen and pelvis. Only ~2% of paragangliomas originate in the chest [3]. In addition, PPGLs typically secrete excess catecholamines (dopamine, epinephrine, or norepinephrine), and patients exhibit a variety of symptoms that overlap with other clinical conditions. Common symptoms linked to catecholamine-secreting PPGLs include hypertension, excessive sweating, headaches, and palpitations. Hypertension is common as paroxysmal and chronic hypertension have been observed in ~95% of patients with PPGLs [4, 5, 6].

In roughly 40% of cases involving PPGLs, a pathogenic variant can be traced back to one of the susceptibility genes, while the remaining 60% are categorized as seemingly spontaneous occurrences i.e. sporadic. These genetic pathogenic variants within PPGLs can be broadly classified into four genetic groups: cluster 1A - *SDHx* genes; cluster 1B - *VHL/EPAS1*; cluster 2 - *kinase signaling (KS)*; and finally, sporadic cases. The most commonly encountered pathogenic variants are found within the *SDHx (A/B/C/D/AF2)* genes. Notably, *SDHx* and *VHL/EPAS1* are characterized by pseudohypoxia, impacting the hypoxia-signaling pathway. The *KS* category encompasses pathogenic variants, such as *RET*, *NF1*, while sporadic cases represent those for which the specific pathogenic variants are yet to be identified [7].

Diagnosing PPGLs is especially important for those patients who express a higher metastasis rate. For example, trunk paragangliomas exhibit metastasis rates as high as 60%, making their detection notably relevant for patient outcomes [8, 9]. Moreover as undiagnosed PPGLs are associated with high morbidity rates [10], it is essential to localize the PPGLs and monitor their progression. For example, catecholamine-induced cardiomyopathy can result in fatal cardiovascular complications in patients [11]. In such cases, early diagnosis holds the promise that the appropriate treatments and interventions can be employed for improved patient outcomes.

Presently, CT is the preferred imaging modality to localize the PPGLs, track their progression, and determine their metastatic potential. However, some intrinsic properties of PPGLs render their detection in CT challenging. For example, they vary in size and range from 1cm - 15cm in maximum diameter. They also vary in their morphology with smaller PPGLs being typically homogeneous, whereas larger PPGLs exhibit central necrosis. Additionally, PPGLs exhibit macroscopic fat, hemorrhage, or calcification that make it difficult to identify and distinguish from other masses, such as adenomas [12, 13, 14]. Finally, they exhibit different biochemical properties based on their genetic cluster. Due to these factors, approximately 25% of patients with PPGLs remain undiagnosed for their lifetime [10].

In clinical practice, tumor size is routinely tracked using the specified RECIST guidelines [15]. However, it is cumbersome to measure the size of each PPGL seen in each slice of a CT volume, and the largest extent of a tumor in one slice of the CT volume is recorded. Furthermore, as the volume of CT exams has been increasing year after year [16], it is cumbersome for radiologists to manually identify PPGLs and measure their sizes in newly acquired CT scans during a busy clinical day. Despite the weak annotations and aforementioned challenges, it is clinically relevant to detect PPGLs as they have the potential to greatly improve patient outcomes. Presently, an expensive genetic test is the only approach towards accurate estimation of the PPGL genetic cluster [17, 18]. An automated tool that can localize PPGLs for sizing in non-invasive CT imaging is of relevance in the radiology workflow.

It is also of clinical relevance for downstream tasks: (1) the rough localization of PPGLs can be considered for the subsequent precise voxel-level annotation and concomitant training of a 3D segmentation model to directly define the tumoral extent, and (2) deriving imaging-based biomarkers from the segmentations to non-invasively identify their genetic makeup [18, 19, 20]. There is little prior work for the task of detection and classification of PPGLs into their genetic clusters. Santra et al. [18] used cropped patches of PPGLs and achieved an F1 score of 46% for the classification of specific genetic cluster types. Their result underscores the difficult nature of the detection and classification of this rare type of cancer.

We propose the first known work to detect PPGLs in contrast-enhanced CT volumes utilizing a segmentation framework. We put forth a cost-effective solution for the generation of weak 3D annotations from prospective 2D radiologist annotations in CT. Using these weak 3D annotations as ground-truth, we trained 2D and 3D nnUNet models to detect PPGLs via a proxy segmentation task. Our state-of-the-art results show that PPGLs can be detected via segmentation, and thereby help in the management of patients with this rare cancer. The remainder of the paper is structured as follows: Section 2 presents the materials and methods, experiments and results are provided in Section 3, and Sections 4 and 5 discuss our findings.

2. Materials and Methods

Patient Sample.

This retrospective study was Health Insurance Portability and Accountability Act (HIPAA) compliant and was approved by the Institutional Review Board at our institution. The requirement for signed informed consent from the patients was waived. The Picture Archiving and Communication System (PACS) at our institution was queried for patients who underwent a CT examination between 1999 and 2022. A total of 300 portal venous phase CT volumes were collected for 289 patients. Certain patients underwent separate CT exams for their head-neck and chest-abdomen-pelvis regions. The acquisition of contrast-enhanced CT volumes followed a consistent protocol involving a fixed delay of 70-seconds post-intravenous contrast material administration. The slice thicknesses ranged from 1 to 10 mm, and a total of 1010 PPGLs were identified.

The focus of this work was to detect PPGLs in the body. For each scan, all PPGLs that were identified belonged to one of four genetic cluster groups: *SDHx*, *VHL/EPAS1*, *KS*, and sporadic though a genetic testing. The 300 volumes contained 150 *SDHx* (50%), 58 *VHL/EPAS1* (~19.3%), 51 sporadic (~13.6%) and 41 *KS* volumes (~17.1%). Volumes containing tumors in the head or neck (n=42) were excluded, yielding 258 volumes (one per patient). We also excluded volumes where no lesions had been observed (n=3). The remaining 255 CT volumes were randomly divided into ~80% training (202) and ~20% test (53) data subsets. The training data subset contained 695 PPGLs and comprised of 90 *SDHx* (~44.5%), 27 *VHL/EPAS1* (~13.4%), 44 *KS* (~21.8%), and 41 sporadic (~20.3%) cluster type volumes. The test dataset contained 153 PPGLs and comprised of 21 *SDHx* (~39.6%), 12 *VHL/EPAS1* (~22.6%), 10 *KS* (~18.9%), and 10 sporadic (~18.9%) cluster type volumes.

PPGL annotation.

In clinical practice, radiologists routinely scroll through the CT volume to identify abnormal findings. The RECIST criterion [15] is used to measure the extent of a tumor and they annotate a few “significant” tumors depending on their size ($\geq 1\text{cm}$). As many tumors can be found in a CT volume, radiologists measure them on one slice only as doing it across all slices is tedious. We replicated this annotation process for the PPGLs. First, we loaded the CT volume into ITK-SNAP [21] with a soft tissue window center and width of [50, 450] HU. A trained grader visually determined the axial slice containing the maximum diameter of a PPGL and drew a bounding box around each tumor. These 2D bounding boxes mimicked the prospective annotations performed by the radiologists. These annotations were then verified by a senior board-certified radiologist with 25+ years of experience. As the tumors persisted for at least the 3 adjacent slices above and below the current annotation, we extended the 2D bounding box to these adjacent slices. This process generated weak 3D bounding box annotations in a cost-effective and time-efficient manner. CT volume regions that were covered by these 3D bounding boxes were defined as containing PPGLs. The objective was to use these weak annotations to train a segmentation model for PPGL localization in CT volumes for subsequent fine-grained annotation.

Body Region Segmentation.

Since PPGLs typically range from 1cm – 15cm in size, a segmentation model trained directly with the 3D boxes as segmentation masks will exhibit degraded performance due to a class imbalance between the foreground (PPGL) and background (all other regions) class. We addressed this class imbalance issue by using the body region segmentation as shown in Fig. 1 for each CT volume. These segmentations were generated with the TotalSegmentator [22] tool. TotalSegmentator produced segmentation masks for various organs in CT, and it distinguished the body region from the background in the CT. The PPGL 3D bounding box masks were then merged with the body region segmentations to produce ground-truth masks that were used to train the segmentation models.

Detection of PPGLs using nnUNet.

The self-configuring nnUNet segmentation framework [23] was employed to detect PPGLs via a proxy segmentation task. It is currently the *de-facto* standard for segmentation tasks as it has won multiple medical segmentation competitions [23, 24]. The nnUNet model

has been rigorously validated [24] for many tasks, including multi-organ segmentation in CT and MRI [25, 22] and lymph node segmentation [26] among others. It has often outperformed other architectures [24], such as transformer-based approaches [27]. For more details on the model performance, the reader is referred to the original publications [23, 24].

The input CT volumes are pre-processed by the framework during the dataset “fingerprint” assessment phase. Appropriate characteristics for the dataset are automatically determined, which includes intensity normalization and resampling to a consistent spacing among others. Specifically for normalization, the intensity values from the foreground classes (all but the background) were collected for all training cases, and the mean/standard deviation as well as the 0.5 and 99.5 percentile of the values were computed. These values were clipped to the percentiles and followed by subtraction of the mean and division with the standard deviation. The same normalization was applied to each volume in the training data subset (for an input channel). This normalization was suitable for modalities presenting physical quantities, such as CT volumes.

The base network in the framework is a UNet [23, 28], which is automatically optimized for the dataset based on its fingerprint. This includes the determination of optimal hyper-parameters, such as a large batch size for accurate gradient computation, automatic patch size adjustment, and adequate depth of the network to account for a receptive field as large as the patch size. Other parameters, such as the number of epochs for training, were set to default values, but they can be customized based on requirements. In this work, only the default parameters for training were used.

Due to the comparable performance to prior works, two nnUNet models (2D and 3D full-resolution) were trained to detect the PPGLs via a proxy segmentation task. During training of both nnUNet models, our pipeline took a CT volume and its corresponding ground-truth mask as input. As previously mentioned, the CT volume was automatically normalized. The segmentation mask contained 3D bounding boxes that roughly delineated the location of the PPGLs in the body as well as the body region mask. If the input CT volume was resampled according to the dataset fingerprint, then the corresponding mask was resampled as well.

The models learned to generate a segmentation for the CT volume and iteratively refined it via a loss function (L), which was a combination of the dice loss and the binary cross entropy loss [29] as shown in Eq. 1. The cross-entropy loss (C) asserts equal learning amongst all voxels in the volume as it evaluates the predictions for each voxel and averages them across the volume. However, for unbalanced classes (body region mask vs. PPGLs), the use of cross-entropy can pose a problem as the loss is dominated by the prevalent class (body region mask). On the other hand, the Dice loss (D) promotes class balancing through the computation of the overlap between the ground-truth (p) and prediction ($\hat{p}$). This loss function places an equal emphasis on all classes in the ground-truth mask and penalizes low-confidence predictions.

$$L = C(p, \hat{p}) + D(p, \hat{p}) \quad (1)$$

$$C(p, \hat{p}) = -[p \log(\hat{p}) + (1 - p) \log(1 - \hat{p})] \quad (2)$$

$$D(p, \hat{p}) = 1 - \frac{2 \sum p \hat{p}}{\sum p + \sum \hat{p}} \quad (3)$$

3. Experiments

Implementation.

Following [23], both the 2D and 3D nnUNet models were trained using 5-fold cross-validation for a total of 1000 epochs (default). To optimize the network weights, stochastic gradient descent with Nesterov momentum ($\mu = 0.99$), an initial learning rate of 10^{-3} , and a batch size of 1 was utilized. These parameters were automatically configured by the nnUNet framework with the help of the dataset fingerprint. A learning rate decay strategy following the “poly” learning rate policy [30] was employed. The models learned to generate a segmentation for the CT volume and iteratively refined it with a combination of binary cross-entropy and soft Dice losses as described in Eq 1. For each output from the model, the corresponding ground truth segmentation mask was used for calculating the losses. The objective of the training was to minimize the summation of the losses (L) considering multiple resolutions, i.e., $L = w_1 \times L_1 + w_2 \times L_2 + \dots$, where $w_2 = \frac{1}{2} \times w_1$, $w_3 = \frac{1}{4} w_1$ and so on, such that $\sum_i w_i = 1$. A variety of data augmentations were implemented following the original implementation [23]. All experiments were done on a workstation running Ubuntu 22.04 LTS with an NVIDIA Tesla V100 GPU.

Baselines.

The first experiment that was conducted was to compare the performance of the two configurations of the nnUNet model: 2D nnUNet vs. 3D full-resolution nnUNet. As previously described in Section 2, the nnUNet models have achieved superior performance across a variety of segmentation tasks [23, 24] compared to other methods [27] and the base architecture of the model is a UNet [28]. Next, we noticed that there were many small predictions made by both nnUNet models that were not tied to the main connected component during a qualitative assessment.

To evaluate the degree of model performance when small predictions were excluded, we designed four experiments. 1) All predictions from the 2D nnUNet model were used for metric computation; 2) A size threshold of 250 voxels was set to exclude any small predictions by the same 2D nnUNet model; 3) All the lesions predicted by the 3D nnUNet model were used for metric computation; 4) Finally, a size threshold of 250 voxels was applied to exclude small predictions from the same 3D nnUNet model. We carried out all these experiments considering all lesions together (overall) irrespective of any genetic cluster information. Next, we repeated the same set of experiments for the test CT volumes

in each genetic cluster. The threshold of 250 voxels was selected after connected component analysis (CCA) was run on the model predictions. The number of voxels in each component was determined and categorized into four bins: 0 – 500 voxels, 500 – 2000 voxels, 2000 – 10000 voxels, 10,000+ voxels. Most of the smaller predictions fell into the 0–500 voxels bin. After a qualitative assessment of the predictions in this bin, the threshold of 250 voxels was chosen to eliminate the false positives.

Additionally, an ablation experiment was conducted to evaluate the performance of the best segmentation model without the use of the body region mask. The purpose was to assess the benefit of using the body region mask to combat class imbalance (background vs. PPGLs) during model training. Finally, the effect of using 3 adjacent slices above and below the weak PPGL annotation in contrast to using a different number of adjacent slices (1-, 2-, and 4- slices) was also determined. Experiments to determine the right loss function to use were unnecessary as it was empirically found in previous investigations [23, 29] that the combination of the dice and cross-entropy losses improved training stability and segmentation accuracy.

Metrics.

We used precision = $\frac{TP}{TP+FP}$ and recall = $\frac{TP}{TP+FN}$ as the evaluation metrics to quantify the performance of our 2D and 3D nnUNet models. True positives (TP) were those ground truth lesions that overlapped with predicted lesions. False positives (FP) were those predictions that did not overlap with any ground truth. Finally, false negatives (FN) were ground truth lesions that were missed by our model. Dice coefficient was not the appropriate choice of a metric as we detected PPGLs via a proxy segmentation task, and we do not have fine-grained segmentation labels for the PPGLs.

Statistical Analysis.

Statistical analysis of the results was performed using the bootstrapping method proposed by [31, 32, 33] for computer-aided detection. We used the area under the Free-response Receiver Operating Characteristic (FROC) curve (A_z) for comparison. 47 patients were randomly sampled with replacement from the test dataset, and the two detection methods being compared were run on the sample to generate two FROC curves. The difference between the area under the curves ∇A_z was estimated. The bootstrapping method was run 1000 times and generated 1000 values for ∇A_z . The p -values were defined as the fraction of values that were negative or zero. Any difference in performance was considered significant if $p < 0.05$.

4. Results

Overview.

The overall performance of the nnUNet models is tabulated in Table 1 by considering two different thresholds on size of the predictions. A few example results of PPGL detection with the nnUNet models are illustrated in Figures 2 and 3. The examples established the efficacy of the proposed approach in detecting PPGLs from CT volumes. Table 2 tabulates patient-level PPGL detection results in terms of mean, standard deviation, median, and

Interquartile Range (IQR) for both models at the two thresholds. Genetic cluster-wise PPGL detection results of the nnUNet models are provided in Table 3. The breakdown of the statistics (mean, standard deviation, median, IQR) of the patient-level results are shown in Table 4. Table 5 provides details on the ablation experiments conducted in this work.

Quantitative comparison of 2D vs. 3D nnUNet.

As seen in Table 1, for the 153 identified PPGLs in the 53 test volumes, with no prediction volume size threshold, the 3D nnUNet achieved a significantly higher ($p = 0.017$) precision (62.4% vs. 39.3%) and recall (64.1% vs. 27.5%) in contrast to the 2D nnUNet model. With a prediction volume size threshold of 250 voxels, the 3D model's precision increased from 62.4% to 70.0% without a decrease in recall. Moreover, it achieved a significantly higher ($p = 0.009$) precision (70.0% vs. 52.7%) and recall (64.1% vs. 25.5%) compared to the 2D model with the 250 voxel threshold. For the 3D model, the small predictions excluded by the threshold were exclusively false positives. For the 2D nnUNet, the excluded predictions were false positives and false negatives.

From Table 2, the 3D nnUNet model also outperformed the 2D model at a patient level; with the 250 voxels threshold, the 3D model exhibited a higher median precision (84.9% vs. 50.0%) and recall (100.0% vs. 0.0%) than the 2D model. Additionally, from Table 3, the 3D model also outperformed the 2D model at a genetic-cluster level. With the 250 voxels threshold, the 3D nnUNet achieved higher precision for sporadic (72.7% vs. 68.6%), VHL/EPAS1 (63.6% vs. 50.0%), and SDHx (73.1% vs. 66.7%) PPGLs. It also attained a higher recall for all genetic clusters. From Table 4, a median precision of at least 70.8% for all four genetic cluster datasets was achieved with the 3D nnUNet and 250 voxels threshold. In terms of recall, the model had $\geq 90\%$ sensitivity in detecting sporadic, kinase, and VHL/EPAS1 genetic clusters. However, the sensitivity languished at 50% for the SDHx cluster. The 3D nnUNet was not sensitive enough to capture SDHx PPGLs, but it was very confident when it did with a mean and median precision of $\geq 68\%$ and $\geq 82\%$. Meanwhile, the 2D nnUNet's median precision and recall was at or below 50% for three out of four genetic cluster datasets.

Results of Ablation Experiments.

Table 5 details the results of the different ablation experiments. First, the effect of using the body region mask as an additional segmentation class was evaluated. The best performing 3D full-resolution nnUNet (with 250 voxel size threshold) was used, and it was trained with 3D PPGL bounding boxes derived from 3 adjacent slices above and below the weak annotation. As seen from Table 5, the PPGL detection performance of the model trained with the body region mask was better ($p = 0.014$). The results indicated that the performance of the model degraded when the body region mask was removed, which is corroborated by the additional false positive and false negative predictions.

Next, the effect of using a different number of adjacent slices to create the 3D PPGL bounding boxes was determined. The network trained with 3D bounding boxes derived from 1- and 2- adjacent slices above and below the annotation (total of 3 and 5 slices respectively) fared the worst, but there was no significant difference between the model trained with 3D

bounding boxes derived from 3 slices vs. 4 slices ($p = 0.79$). This indicated that either 3- or 4- adjacent slices could be used to create the 3D PPGL bounding boxes and the detection performance was comparable.

5. Discussion & Conclusion

The weak 3D box annotations introduced in this work were sufficient to train a 3D nnUNet model for detecting PPGLs via a proxy segmentation task. The model was successful when a prediction size threshold of 250 voxels was set. Collectively, the results from our findings suggest that the 3D nnUNet can detect different PPGL genetic clusters in our test dataset. Once they are roughly identified in newly acquired CT volumes, precise delineations of the PPGLs can potentially be obtained. However, the same weak annotations were insufficient to train a 2D nnUNet model as it was unable to achieve a similar level of performance. We believe that the 3D context provided by the 3D nnUNet is crucial as the genetic cluster types of PPGLs are not easily distinguished as shown in prior work [18].

Contrasted against the 46% F1 score attained by Santra et al. [18], our F1 score was 66.9% across all the genetic cluster types. The results from Table 3 indicated that the 3D nnUNet model detected PPGLs of all genetic clusters with reasonable precision. This type of cancer is rare and a patient generally has only one genetic cluster type of PPGLs. This is of importance as there were wide disparities in the training data quantity across the clusters. As there were more SDHx PPGL volumes in the dataset in contrast to the other cluster types, the 3D nnUNet model was able to achieve the highest precision for the SDHx genetic cluster.

The main limitation in this work is related to the generation of the ground-truth data. A consequence of using the weak 3D box-based PPGL annotations was the over-estimation (for small tumors) and under-estimation (for large tumors) of the true 3D tumor extent. As seen in Fig. 3, the detection of the PPGLs can be achieved through perfect and imperfect segmentations. For example, in Fig. 3(b), the 3D nnUNet segmented the contour of the left adrenal PPGL of the VHL/EPAS1 cluster type almost perfectly, which contrasts with the approximate bounding box segmentation of the right adrenal tumor of Kinase cluster type in Fig. 3(a). Moreover, from Figs. 3(c) and 3(d), the model sometimes correctly detected tumors, but failed to correctly capture the full extent of the tumor (either with a bounding box or precise contour). Additionally, it struggled to segment large tumors (10,000+voxels) as seen in Fig. 3(d) as there were very few large tumors (17/695) in the dataset. Since it is very challenging to identify the PPGLs in CT volumes [14, 18], the proposed approach can be useful for the rough tumor localization.

Another limitation is the observation that the model inherited the biases of the ground-truth annotation that consisted of weak 3D bounding boxes. For example, it was noticed in certain instances that the prediction would disappear after 7 slices despite the tumor persisting for additional slices. Despite this, the model can help bridge the gap between automated PPGL detection and the subsequent precise annotation of the tumors for re-training a segmentation model to directly segment the entire extent of PPGLs. As an expensive genetic test is currently the only method to precisely determine the genetic cluster of PPGLs, biomarkers

(morphological and functional) derived from CT imaging through the PPGL segmentation have the potential to offer timely personalized patient therapy.

In summary, our proposed approach is the first known work to detect PPGLs in contrast-enhanced CT volumes utilizing a segmentation framework. We used weak 3D annotations to train both 2D and 3D nnUNet models for detecting PPGLs via a proxy segmentation task. The 3D nnUNet model was better than the 2D model ($p < 0.05$) when a prediction size threshold of 250 voxels was used. Compared against prior work, our results are promising for the detection and subsequent classification of the different PPGL genetic cluster types.

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Highlights

1. In this pilot work, we present the first approach to detect Pheochromocytomas and Paragangliomas (PPGLs), which are rare adrenal and extra-adrenal tumors that have the potential for metastasis.
2. An in-house dataset of 255 patients was collected and patients underwent CT of the chest, abdomen, and pelvis. 2D measurements of tumors in the volumes were converted into weak 3D bounding box annotations.
3. Two different nnUNet segmentation models were subsequently trained to detect PPGLs via a proxy segmentation task.
4. Our 3D model achieved a detection precision of 70% and sensitivity of 64.1% respectively, while outperforming a 2D model trained for the same task ($p < 0.05$).
5. Our results represent the state-of-the-art for the challenging task of automated PPGL detection.

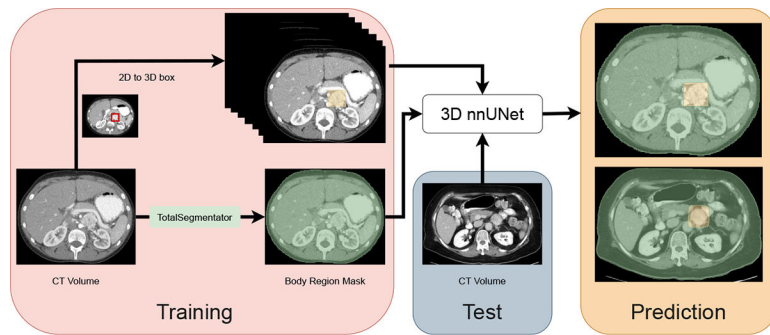


Figure 1:

Framework for the detection of pheochromocytomas and paragangliomas (PPGLs) via a proxy segmentation task using a 3D full-resolution nnUNet. 2D bounding boxes (red box) were formed from the prospectively marked PPGLs in the CT volumes, and these were converted into weak 3D segmentations (yellow box) by propagating them to adjacent slices (± 3). The body region mask (green) from TotalSegmentator was also merged with these weak 3D annotations. The 3D model was trained to predict the corresponding regions in the CT volume where the PPGLs were observed. At test time, the model received a 3D CT volume and detected PPGLs in it.

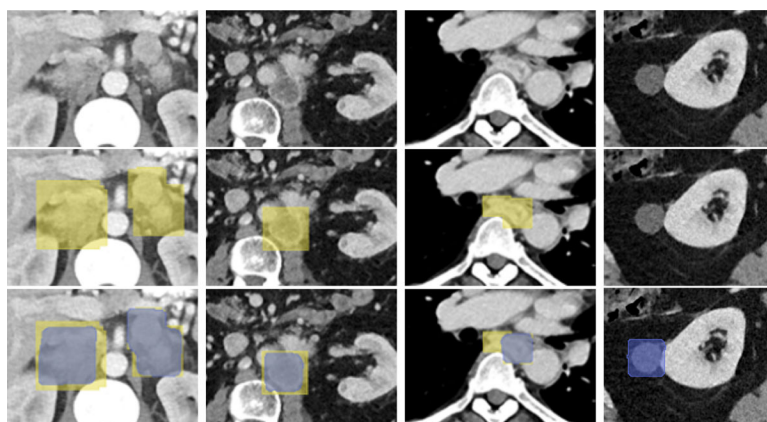


Figure 2:

Rows 1, 2, and 3 show cropped CT slices, ground-truth PPGLs (yellow), and detected PPGLs (blue) overlaid, respectively. Columns 1, 2, and 3 show true positives; a small part of an oblong tumor in column 3 was detected. In Column 4, a false positive is shown.

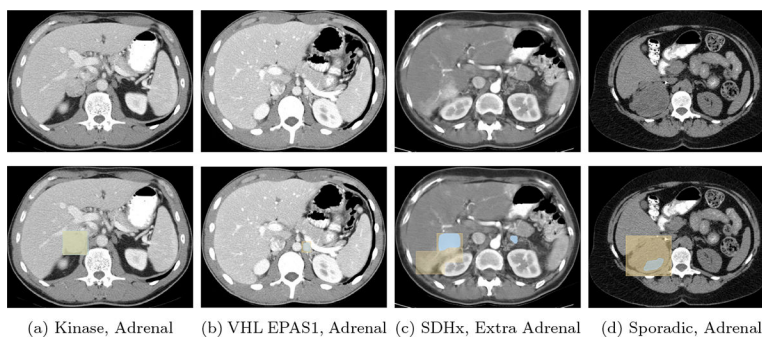


Figure 3:
Qualitative results of the 3D full-resolution nnUNet for detection of adrenal and extra-adrenal PPGLs via a proxy segmentation task for each genetic cluster type.

Table 1:

Overall results of PPGL detection with the different nnUNet models for different prediction size thresholds.
Bold font indicates best results.

Size Threshold (in voxels)	# GT	# TP	# FP	# FN	Precision	Recall
2D nnUNet						
0	153	42	65	111	39.3%	27.5%
250	153	39	35	114	52.7%	25.5%
3D nnUNet						
0	153	98	59	55	62.4%	64.1%
250	153	98	42	55	70.0%	64.1%

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Table 2:
Patient-level PPGL detection results for different prediction size thresholds. Bold font indicates best results.

Metric	Size Threshold 0				Size Threshold 250			
	Mean	Std. Dev.	Median	IQR	Mean	Std. Dev.	Median	IQR
2D nnUNet								
Precision	48.2%	43.3%	50.0%	0.0% – 100.0%	53.7%	44.6%	50.0%	0.0% – 100.0%
Recall	34.7%	43.1%	0.0%	0.0% – 80.0%	30.9%	42.6%	0.0%	0.0% – 66.7%
3D nnUNet								
Precision	62.6%	37.0%	66.7%	33.3% – 100.0%	68.9%	36.0%	84.9%	47.5% – 100.0%
Recall	66.7%	40.2%	100.0%	50.0% – 100.0%	66.7%	40.2%	100.0%	50.0% – 100.0%

Table 3:
Genetic cluster-level detection results at different prediction size thresholds. Bold font indicates best results.

Size Threshold (in voxels)	Genetic cluster	# ground-truth	# TP	# FP	# FN	Precision	Recall
2D nnUNet							
0	Kinase	24	13	12	11	52.0%	54.2%
	Sporadic	39	6	28	33	17.6%	15.4%
	VHL/EPAS1	29	6	11	23	35.3%	20.7%
	SDHx	61	17	14	44	54.8%	27.9%
250	Kinase	24	13	6	11	68.2%	54.2%
	Sporadic	39	4	9	35	30.8%	10.3%
	VHL/EPAS1	29	5	10	24	33.3%	17.2%
	SDHx	61	17	10	44	63.0%	27.9%
3D nnUNet							
0	Kinase	24	15	8	9	65.2%	62.5%
	Sporadic	39	24	11	15	68.6%	61.5%
	VHL/EPAS1	29	21	21	8	50.0%	72.4%
	SDHx	61	38	19	23	66.7%	62.3%
250	Kinase	24	15	7	9	68.2%	62.5%
	Sporadic	39	24	9	15	72.7%	61.5%
	VHL/EPAS1	29	21	12	8	63.6%	72.4%
	SDHx	61	38	14	23	73.1%	62.3%

Table 4:
Breakdown of patient-level PPGL detection results for different prediction size thresholds. Bold font indicates best results.

		Precision				Recall			
Size Threshold (in voxels)	Genetic Cluster	Mean	Std Dev.	Median	IQR	Mean	Std Dev.	Median	IQR
2D nnUNet									
0	Kinase	75.6%	35.4%	100.0%	50.0% – 100.0%	63.8%	41.1%	73.3%	35.4% – 100.0%
	Sporadic	39.8%	42.3%	29.5%	6.8% – 62.5%	40.4%	45.6%	26.8%	0.0% – 87.5%
	VHL/EPAS1	43.8%	41.7%	50.0%	0.0% – 62.5%	29.2%	40.3%	0.0%	0.0% – 54.2%
	SDHx	37.2%	46.2%	0.0%	0.0% – 100.0%	21.3%	39.9%	0.0%	0.0% – 21.3%
250	Kinase	85.2%	33.8%	100.0%	100.0% – 100.0%	63.8%	41.1%	73.3%	35.4% – 100.0%
	Sporadic	36.1%	38.6%	33.3%	4.17% – 50.0%	30.4%	48.1%	0.0%	0.0% – 75.9%
	VHL/EPAS1	42.9%	45.0%	50.0%	0.0% – 75.0%	20.8%	34.2%	0.0%	0.0% – 37.5%
	SDHx	44.5%	47.2%	40.0%	0.0% – 100.0%	21.3%	39.9%	0.0%	0.0% – 14.3%
3D nnUNet									
0	Kinase	71.7%	38.5%	100.0%	37.5% – 100.0%	77.2%	36.4%	100.0%	70.0% – 100.0%
	Sporadic	67.5%	34.3%	70.8%	50.0% – 100.0%	80.4%	34.6%	100.0%	65.2% – 100.0%
	VHL/EPAS1	57.4%	40.1%	63.3%	25.6% – 100.0%	71.4%	38.1%	91.7%	55.4% – 100.0%
	SDHx	58.7%	37.4%	66.7%	33.3% – 100.0%	52.6%	43.5%	50.0%	0.0% – 100.0%
250	Kinase	73.3%	37.0%	100.0%	50.0% – 100.0%	77.2%	36.4%	100.0%	70.0% – 100.0%
	Sporadic	71.2%	35.9%	89.5%	50.0% – 100.0%	80.4%	34.6%	100.0%	65.2% – 100.0%
	VHL/EPAS1	63.8%	38.7%	70.8%	38.3% – 100.0%	71.4%	38.1%	91.7%	55.4% – 100.0%
	SDHx	68.5%	36.5%	82.1%	45.8% – 100.0%	52.6%	43.5%	50.0%	0.0% – 100.0%

Table 5:

Ablation results for PPGL detection with the 3D full-resolution nnUNet model and the 250 voxel size threshold to remove false positives. Bold font indicates best results.

# Adjacent Slices	Body Mask	# GT	# TP	# FP	# FN	Precision	Recall
1	Yes	153	81	53	72	60.4%	52.9%
2	Yes	153	85	49	68	63.4%	55.6%
3	No	153	79	54	74	59.4%	51.6%
4	Yes	153	96	45	57	68.1%	62.7%
3	Yes	153	98	42	55	70.0%	64.1%